

SANJEEV KUMAR

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EDUCATION

Delhi Technological University (Formerly DCE)

B.Tech in Computer Science & Engineering, Minor in Machine Learning

Yuva Shakti Model School

Class XII (CBSE): **90.8%** | Class X (CBSE): **93.2%**

New Delhi, India

Nov 2022 – Jun 2026

Delhi, India

2020 – 2022

EXPERIENCE

Outlier AI

AI Training Specialist & QA Reviewer

Remote

Jan 2026 – Apr 2026

- Evaluated **500+** **LLM-generated responses** across coding, mathematics, and statistical reasoning for correctness, instruction adherence, and reasoning quality using structured evaluation rubrics.
- Performed **error analysis** on model failures and edge cases, identifying and documenting recurring logical, factual, and instruction-following issues to improve evaluation and QA consistency.

The Sparks Foundation

Data Science & Business Analytics Intern

Remote

Feb 2025 – Mar 2025

- Performed **EDA and data cleaning** using Python, Pandas, and NumPy, handling missing values and outliers to identify inventory trends, low-turnover categories, and higher holding costs.
- Built interactive **Tableau dashboards** to track business KPIs, inventory trends, and product-level performance for data-driven analysis.

PROJECTS

GRACE - Efficient Multimodal Hateful Meme Detection | *PyTorch, CLIP, LLaVA, Cross-Attention*

[GitHub](#)

- Contributed to a multimodal classification framework using **PyTorch, CLIP-based image/text representations, asymmetric dual-path cross-attention, and offline LLM-rationale distillation** for efficient image-text reasoning.
- Evaluated the system across Hateful Memes Challenge, HarMeme, and PrideMM, achieving **83.51%, 93.14%, and 85.72% AUROC**, respectively.
- Performed **latency benchmarking, ablation studies, hyperparameter analysis, and failure analysis**; the final system achieved **21 ms inference per meme on a single NVIDIA T4 GPU**.

LLM Inference & Serving Benchmarking System | *Python, PyTorch, vLLM, CUDA, Docker*

[GitHub](#)

- Built an asynchronous **LLM inference benchmarking framework** using vLLM to measure TTFT, TPOT, throughput, p95/p99 latency, GPU utilization, and VRAM across concurrent serving workloads.
- Benchmarked **continuous batching, KV/prefix caching, mixed precision, and quantization** across varying concurrency and context lengths, profiling latency-throughput-memory trade-offs and investigating GPU OOM failures.

Hybrid Document Search & QA System | *Python, FAISS, BM25, LangChain*

[GitHub](#)

- Built a hybrid **retrieval-augmented generation pipeline** combining FAISS dense retrieval with BM25 sparse retrieval for document-grounded question answering.
- Benchmarked retrieval across **10+ document sets** and reduced hallucination rate by **30%** versus naive prompting through experiments with chunking, retrieval, and context selection.

TECHNICAL SKILLS

Languages: C++, Python, Java, TypeScript, JavaScript, SQL

Backend & Databases: FastAPI, Node.js, Express.js, PostgreSQL, MySQL, MongoDB, Redis, Prisma, REST APIs

AI/ML: PyTorch, TensorFlow, Scikit-learn, Transformers, LangChain, LangGraph, RAG, ChromaDB, BM25

Tools & Platforms: Git, Docker, AWS, LangSmith, Streamlit, GitHub Actions, n8n

ACHIEVEMENTS

Competitive Programming: Solved **800+** DSA problems across LeetCode and Codeforces; LeetCode Max Rating **1716** (Top 12%), Codeforces Specialist (**1517**).

Academic: Scored **98.87 percentile** in JEE Main.

Hackathons: Participated in **SignalWatch** and the on-campus **HCL Hackathon**, gaining hands-on experience in collaborative problem solving and rapid prototyping.

Leadership & Volunteering: Volunteered at **Engifest**, DTU's annual cultural festival, supporting event coordination and operations.

Certifications: Andrew Ng's Machine Learning & Deep Learning Specializations (DeepLearning.AI).